

$\left. \begin{matrix} B \rightarrow \text{Bytes} \\ b \rightarrow \text{bits} \end{matrix} \right\}$

- bit = either 0 or 1
- Byte = 8-bit or 8 bits
- KB = $2^{10} \text{ B} = 2^{10} \text{ Bytes}$
- MB = $2^{10} \text{ KB} = 2^{20} \text{ Bytes}$
- GB = $2^{10} \text{ MB} = 2^{30} \text{ Bytes}$
- TB = $2^{10} \text{ GB} = 2^{40} \text{ Bytes}$

• Addressability: In a memory setup of $2^n \times m$, it means $2^n \rightarrow$ diff. locations.

$n \rightarrow$ no. of add. lines
 $m \rightarrow$ length of data

In general Computer ^{memory is} ~~are~~ bit addressable it means $2^n \times 1$
 is ~~are~~ byte addressable it means $2^n \times 8$
 is ~~are~~ word addressable it means $2^n \times 16$ or $2^n \times 32$

it depends on word size.

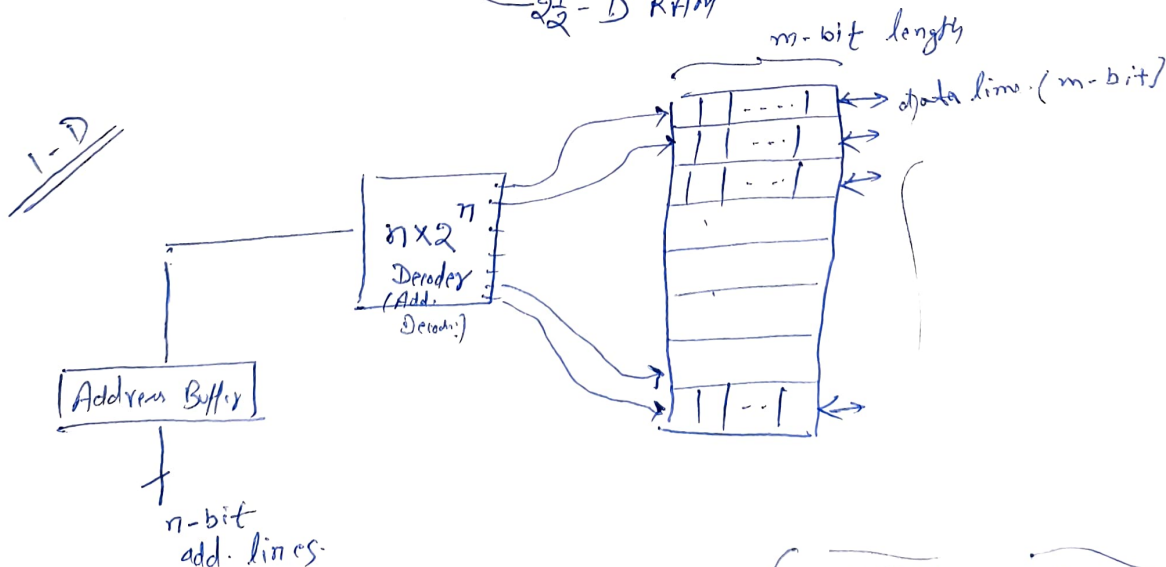
e.g.: 256×8 means 256 diff. locations, each location is capable of storing 8-bit, the add. lines = 8 $\because 2^8 = 256$

$256 \text{ K} \times 8$ means $2^{18} \times 8$, 2^{18} diff. locations, each location is capable of storing 8-bit, no. of add. lines = 18.

RAM organization.

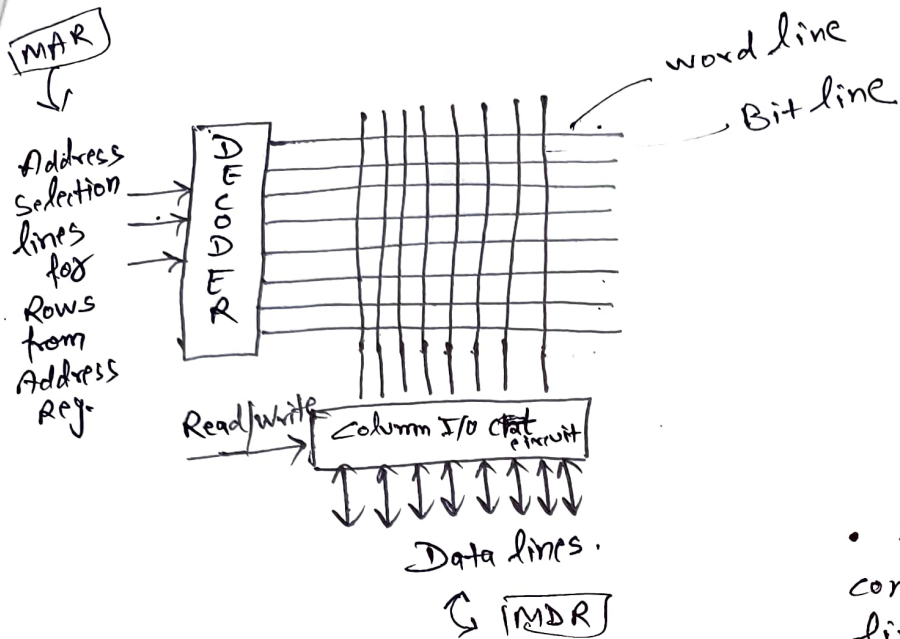
1-D RAM
 2-D RAM
 $2^{1/2}$ -D RAM

$\{2^n \times m\}$
 RAM size



$2 \text{ Mbps} \rightarrow 256 \text{ KBps}$
 2 MBps
 $40 \text{ MB} \rightarrow \text{time} \rightarrow 20 \text{ Sec.}$
 (8) times. 16 sec.

2-D & 2-1/2 D Memory Organization



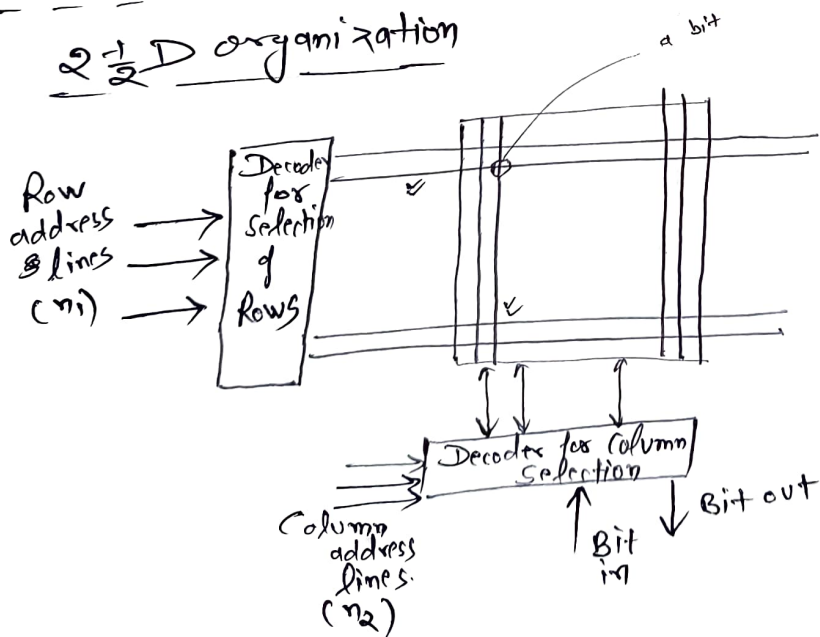
2-D organization

- no 1 Decoder needed $\rightarrow 1$
- if we are having n -bit address then we are having 2^n no. of words line
- No. of Bit lines (columns) are fixed, depending upon the word size.
- Depending upon the input bit combination, one of the output line (word line) is selected.

- No. of Bit lines are not dependent on the Row Decoder size.
- Whether the bits are ~~coming~~^{going} out or coming in that is decided by Read / Write control enable signal.
- if the memory configuration is

then $2^n \times m$
 $2^n \rightarrow$ no. of rows
 $m \rightarrow$ no. of columns.

2-1/2 D organization



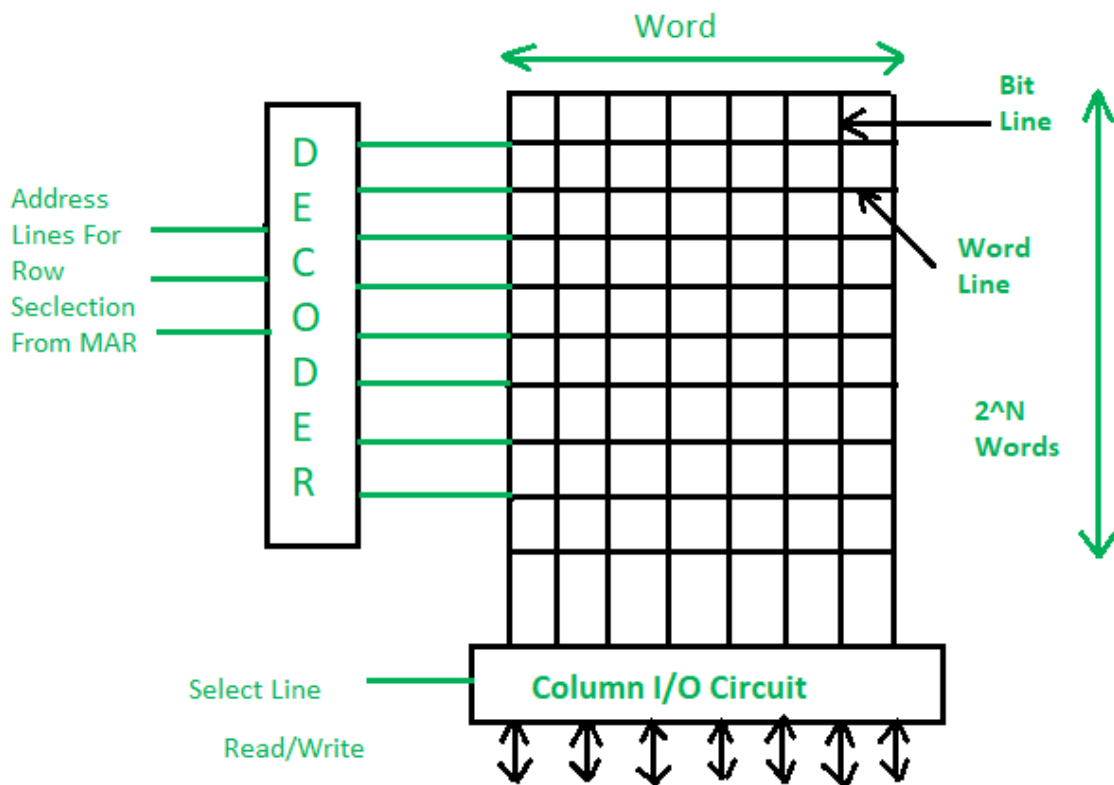
if we are having n -bit address reg. then $(n_1 + n_2 = n)$

- $2^{n_1} \rightarrow$ no. of rows
- $2^{n_2} \rightarrow$ no. of columns.
- Each crossing of rows & columns is representing a single bit
- no. of Decoders needed 2
- if we require 32-bit data then 32-parallel circuit will be working.

2D vs 2.5D Memory Organization

Memory organization plays a crucial role in determining the speed, scalability, and complexity of digital systems. Two common architectures—2D and 2.5D—offer different trade-offs in decoder design and access efficiency.

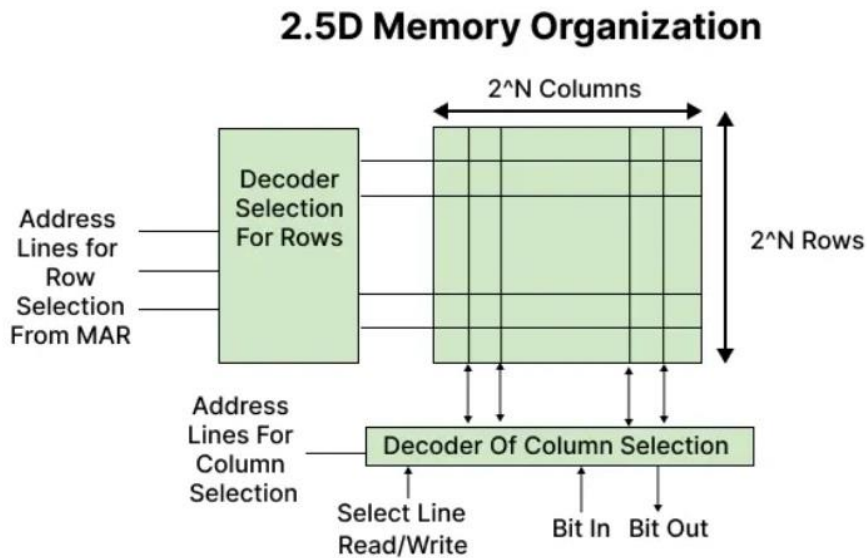
2D Memory Organization



2D Memory Organization

- **Decoder Setup:** Uses a *single decoder* to select both row and column addresses.
- **Access Mechanism:** The decoder interprets address lines and activates one word line corresponding to a memory location.
- **Efficiency:** Simpler design, but slower access due to shared decoding logic.
- **Use Case:** Best suited for small-scale memory systems where simplicity is prioritized.

2.5D Memory Organization



- **Decoder Setup:** Employs *two separate decoders*—one for rows and one for columns.
- **Access Mechanism:** Row decoder selects the word line, while column decoder selects the bit line independently.
- **Efficiency:** Faster access and better scalability due to parallel decoding.
- **Use Case:** Ideal for larger memory systems requiring higher performance and modularity.

Comparison Table

Feature	2D	2.5D
Decoder Setup	Single	Separate
Access Speed	Slower	Faster
Design Complexity	Lower	Higher
Scalability	Limited	Better
Use Case	Small systems	Large systems

Conclusion

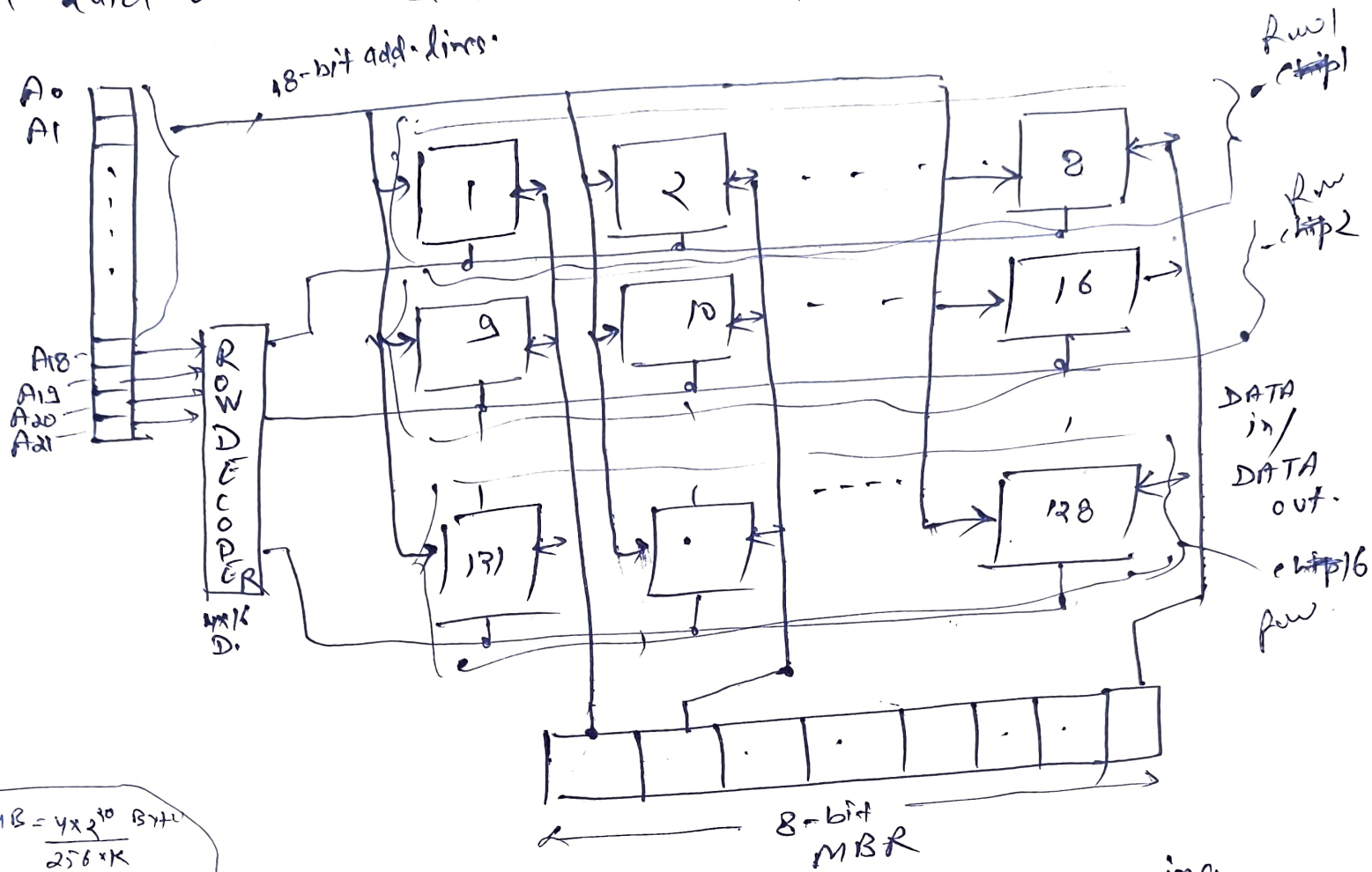
While 2D memory organization offers simplicity and ease of implementation, 2.5D architecture provides enhanced performance and scalability. The choice between them depends on system requirements such as speed, size, and complexity.

Organization of Semiconductor Memory: ($2D \times 2^{1/2}D$) ^{byte add.} (2)

Let us suppose that we want to organize a 4 MB memory using the chips $256K \times 1$. Conceptually, if we put 8 such ~~pin~~ chips in a row, each chip contributing one-bit of a byte. Then we can construct a $256K \times 8$ memory system. This $256K$ memory will need an 18-bit address. The add. lines can be fed as A_0 to A_{17} to each of the 8 chips in a row so as to select one out of $256K$ elements. Since we need 4 MB, so sixteen such rows can be used together to make up the total memory.

A memory with 4 MB capacity will need a 22-bit address, as $2^{22} = 4MB$. The high order 4-bits of the 22-bit add. can be then be used to select one out of 16 rows in the memory organization.

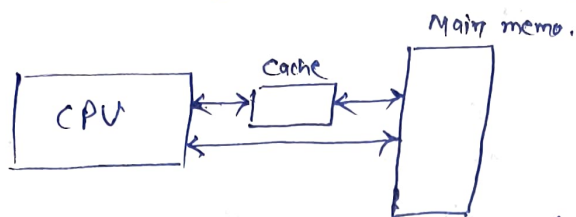
2D
In a 2D organized memory system the chips are laid out as several rows and columns.



$$\begin{aligned}
 4MB &= \frac{4 \times 2^{20} \text{ Bytes}}{256 \times K} \\
 &= 2^{21-18} \\
 &= 2^3 = 8
 \end{aligned}$$

Fig. 2D org. of 4MB mem. using 256K x 1 chips.

Address mapping & Replacement in cache memo.



- CPU communicates with main & cache memory directly
- CPU places the main-memo. address. , so as part of program CPU addresses the main memo. loc. then there should be some mechanism (i.e. mapping) which move in the particular Block in cache.
- In general a word is made up of 2 Bytes
- & A Block which is larger in size is made up of some no. of words.

$\text{bit} < \text{byte} < \text{word} < \text{block} < \text{Page}$

(1) Direct-mapped cache:

- The main memory block can be placed in one and only one place in the cache.

Locations:

Cache:	0	1	2	3	4	5	6	7	8	9
Main:	0	1	2	3	4	5	6	7	8	9
	10	11	12	13	14	15	16	17	18	19
	20	21	22	23	24	25	26	27	28	29
	30	31	32	-	-	-	-	-	-	-

So in the location '0' of cache, it can accommodate content from main loc. 0 or 10 or 20 or 30 - - etc.

Similarly in cache location '2'

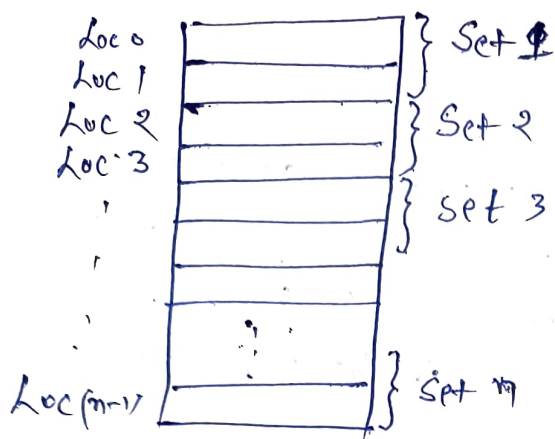
it can accommodate main location 2 or 12 or 22 or 32, ...

formula:

$$\text{Cache address} = (\text{Main Address}) \bmod (\text{no. of cache loc.})$$

e.g. $M.A = 23$, $23 \bmod 10 = 3$

(ii) Set-Associative mapping - (n-way set associative mapping)
 is combination of Direct mapped & fully Assoc. mapping.



it means. the Cache location '0' & '1' can accommodate content of main location

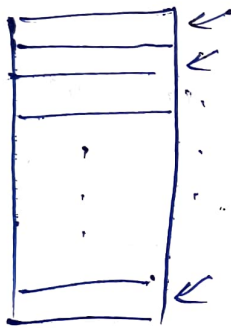
0, 10, 20, 30, 40, ...
 & 1, 11, 21, 31, 41, ...

here $n=2$

so it is 2-way set associative mapping.

(iii) Full-Associative mapping.

Cache loc.



block of main location can go any of the empty location of cache.

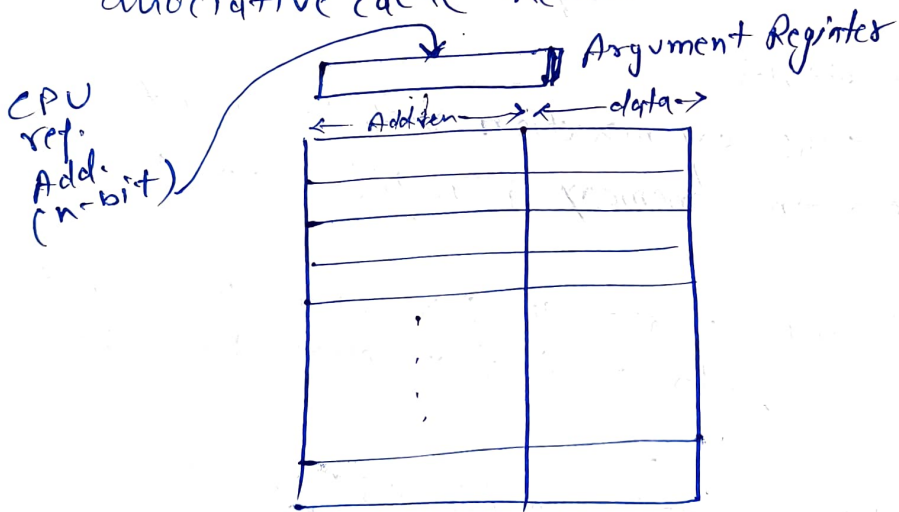
Note :- if any cache location is not empty to accommodate further, the replacement mechanism is required in this case.

Replacement Algo:- When a new Block is brought into the cache, one of the existing blocks must be replaced, by a new block
 there are four most common replacement algorithm

- (i) Least-Recently-Used (LRU), (ii) First-in-First-out (iii) Least Frequently Used (LFU)
- (iv) Random.

Associative memory (Associative cache memo.) (1)
in reference to associative mapping (Fully-ano.)

- The fastest and most flexible cache organization uses an associative memory.
- This associative memory stores both the address and content (data) of the memory word.
- This permits any cache location to store any word from main memory.
- When CPU referenced n -bit address in the Argument Register, then Associative memo. is searched for a matching address. (This matching is sequential)
- if address is found, the corresponding m -bit data is read and sent to the CPU. (Only content is sent to CPU)
- if no matches occurs, the main memory is accessed for the word.
- the address-data pair is then transferred to the associative cache memo.



Note: Associative memories are very much expensive compared to other random-access memories because of the added logic associated with each cell.

Cache Mapping Technique

Cache mapping techniques determine how main memory addresses are mapped to cache locations. The three primary cache mapping techniques are:

1. Direct Mapping:

- In direct mapping, each block of main memory has only one specific cache location where it can be placed.
- Memory blocks are mapped to cache lines using a modulo operation on the block number and the total number of cache lines.
- It is the simplest mapping technique but can lead to higher cache conflicts due to multiple memory blocks mapping to the same cache line.

2. Set-Associative Mapping:

- Set-associative mapping lies between direct and fully associative mapping.
- Memory blocks are grouped into sets, and each set contains multiple cache lines.
- A block from main memory can be placed in any cache line within a specific set.
- This technique reduces conflicts compared to direct mapping but has more flexibility than fully associative mapping.

3. Fully Associative Mapping:

- In fully associative mapping, each block of main memory can be placed in any cache location.
- It doesn't have predefined slots for blocks; rather, it searches the entire cache to find a matching block.
- This method provides maximum flexibility and minimizes conflicts but requires more complex hardware for cache searching.

Cache mapping techniques aim to optimize cache performance by reducing the number of cache misses, conflicts, and access time. Each technique has its advantages and trade-offs in terms of hardware complexity, efficiency, and performance. The choice of mapping technique depends on the specific requirements of the system and the balance between hardware complexity and cache performance.

Virtual Memory Concept

Virtual memory is a memory management technique that provides an abstraction layer between the programs running on a computer and the physical memory (RAM). It allows a system to execute programs that may be larger than the available physical memory by temporarily transferring data from RAM to secondary storage.

Imagine you're working on your computer. You have a lot of different programs running, like a web browser, a word processor, and a music player. Each program needs space in the computer's memory (RAM) to run and store information.

Now, what if your computer doesn't have enough memory to handle all these programs at once? That's where virtual memory comes in.

Virtual memory is like a backup memory system on your computer. It helps manage the available space in your RAM. When your computer's physical memory (RAM) gets full because of all the running programs, virtual memory steps in.

It temporarily uses part of your computer's hard drive as an extension of the RAM. It takes pieces of data that aren't immediately needed by programs and stores them on the hard drive, freeing up space in the RAM for the more important and actively used data.

So, even if your computer's RAM is full, virtual memory helps keep things running smoothly by moving some data back and forth between the RAM and the hard drive as needed.

Address Mapping:

- Address mapping is the process of determining where data should be stored or retrieved within a memory system.
- In the context of caches, it involves mapping memory addresses to specific locations within the cache. Various techniques like direct mapping, set-associative mapping, or fully associative mapping are used to determine where specific data from main memory should be placed within the cache.
- In virtual memory systems, address mapping translates virtual memory addresses used by programs into physical memory addresses where the data is stored in RAM or on disk.

Replacement Techniques

- Replacement comes into play when the memory space, whether it's a cache or virtual memory, is full and new data needs to be brought in.
- In cache memory, when a new piece of data needs to be loaded into a cache line but that line is already occupied, a decision has to be made about which data currently in the cache should be replaced with the new data. This decision is based on a replacement policy (like LRU - Least Recently Used, FIFO - First In First Out, etc.).
- In virtual memory, if the physical memory (RAM) is full and a new page needs to be brought in, the operating system needs to decide which existing page in the RAM should be replaced and moved to disk. Again, replacement policies are employed here to determine which page to swap out.

What is Virtual Memory?

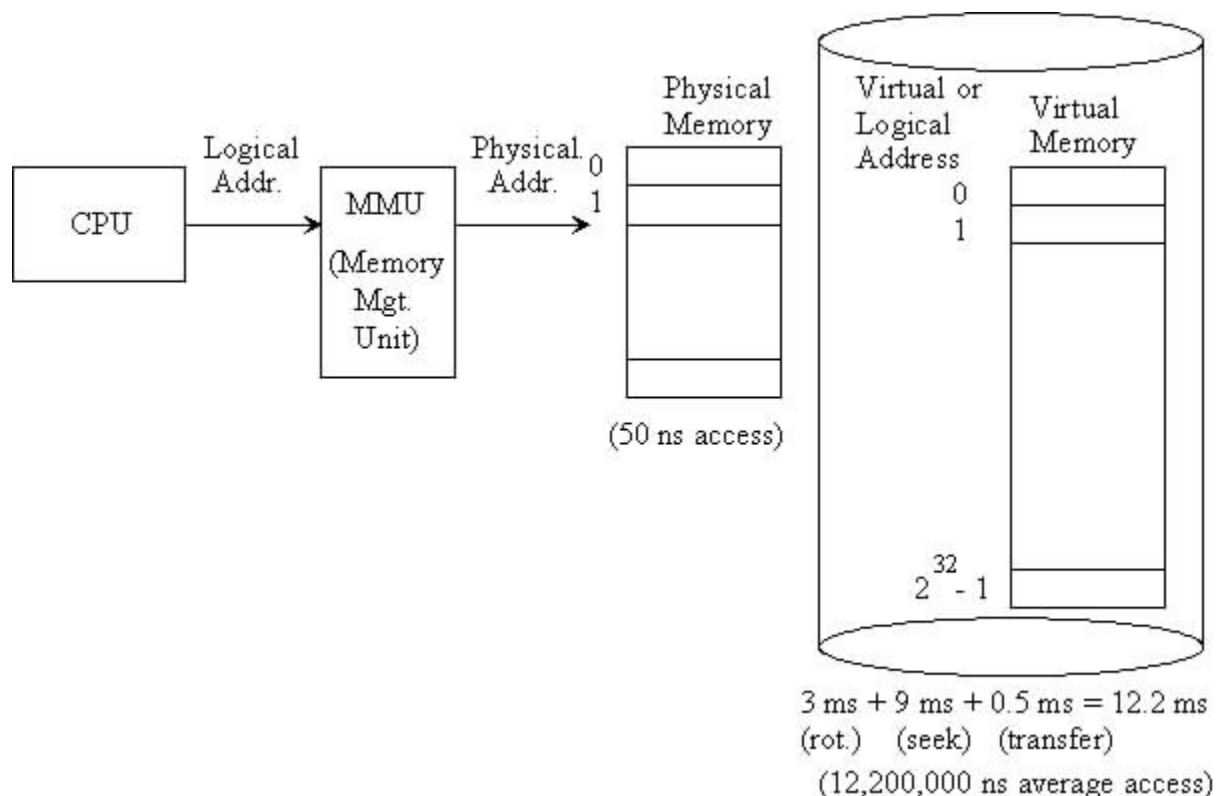
Virtual memory is a memory management technique used by an operating system (OS) to make the **computer appear to have more memory (RAM)** than it actually has.

It does this by using a portion of the **hard disk (secondary storage)** as if it were an extension of the **main memory (RAM)**.

$$\text{Virtual memory} = \text{RAM} + \text{Swap Space (on disk)}$$

Why Virtual Memory is Needed

- RAM is limited and expensive.
- Large programs may not fit entirely in RAM.
- It allows **multiple programs** to run simultaneously (multitasking).
- Prevents “out of memory” errors.

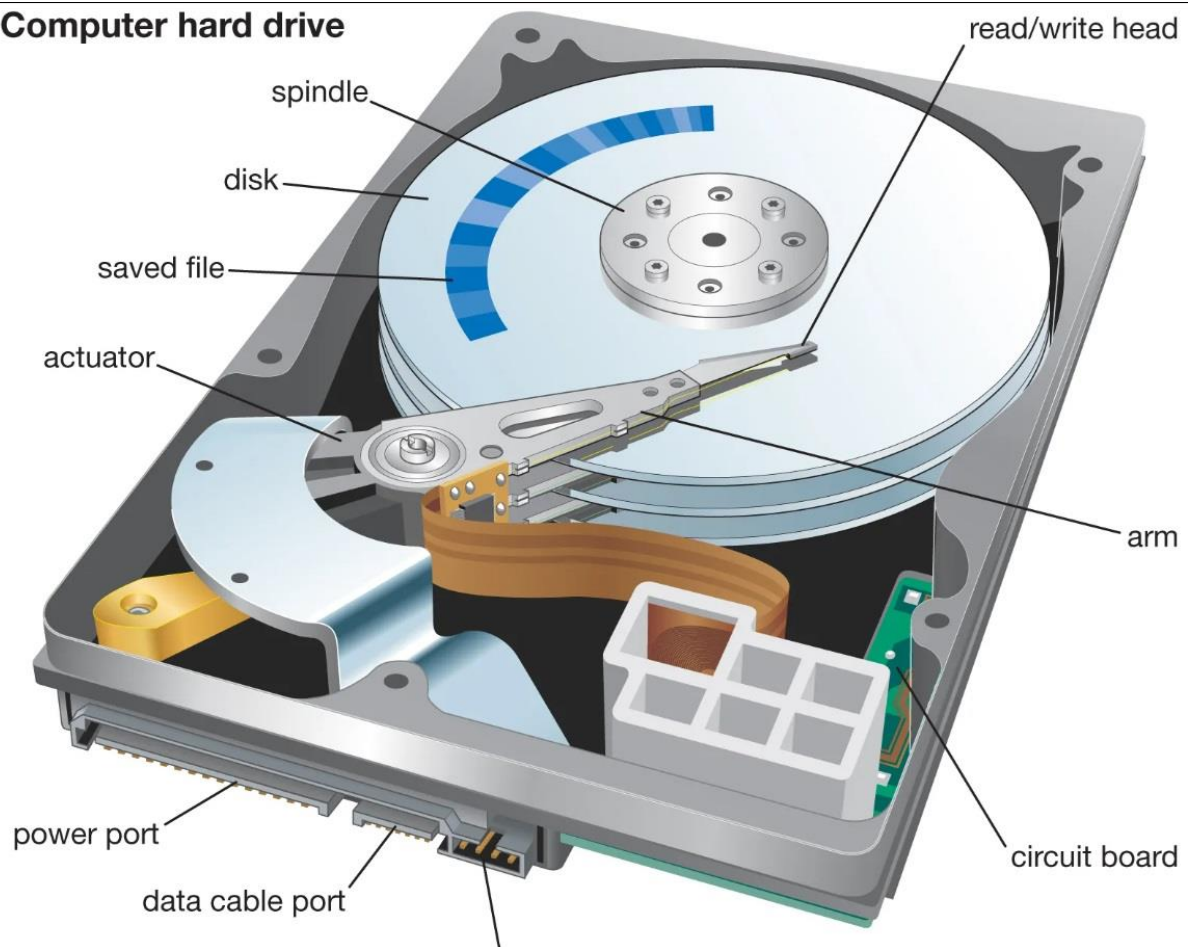


When fetching a page during a page fault:

Total Disk Access Time i.e.

$$T_{\text{disk}} = T_{\text{seek}} + T_{\text{rotation}} + T_{\text{transfer}}$$

Computer hard drive



Seek Time (T_s)	Time for the disk's read/write head to move to the correct track .	2–10 milliseconds
Rotational Latency / Rotation Time (T_r)	Time waiting for the desired sector to rotate under the head.	~4 ms (average for 7200 RPM disk)
Transfer Time (T_t)	Time to actually read or write the data once the head is in position.	< 1 ms for 4 KB page

How It Works

1. **Programs are divided into pages** (usually 4 KB each).
 2. **Only the required pages** of a program are loaded into RAM.
 3. The rest remain stored on the **hard disk** in a special area called the **swap space** or **page file**.
 4. When a page needed by the CPU is **not in RAM**, a **page fault** occurs.
 5. The OS then loads that page from disk into RAM.
-

Key Components

Term	Description
Page	Fixed-size block of data stored in virtual memory.
Frame	Block of physical memory (RAM) where a page is loaded.
Page Table	Keeps track of which virtual pages are stored in which physical frames.

Swap Space / Page File Disk area used as virtual memory extension.

Example

Suppose:

- System has **4 GB RAM**
- Virtual memory size is **6 GB**

Then, the OS can **run programs totalling up to 8 GB**, by keeping 4 GB in RAM and the remaining 4 GB on the disk (temporarily swapped in/out as needed).

Note- balance between usability and performance is important

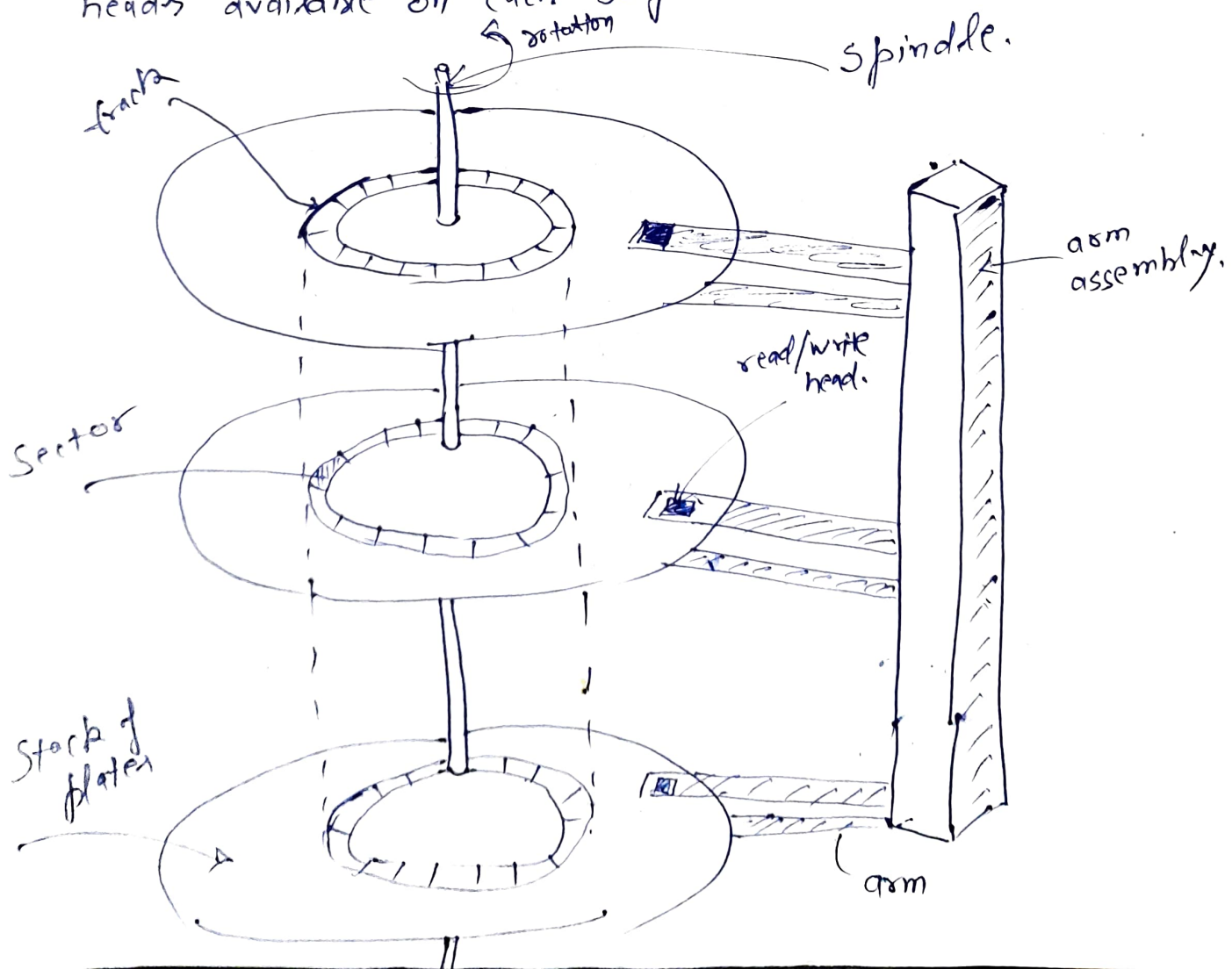
* Optical Disc : A storage medium from which data is read and written by lasers. Optical ~~discs~~ ^{(3) discs} can store much more data - up to 6 GB more than most portable magnetic media such as floppies.

there are three basic types of optical disks

- (i) CD-ROM (Read-only) [Do not allow us to burn content i.e. No writing for user]
- (ii) WORM (Write-Once-Read-Many) [we can burn the content on disc once]
- (iii) EO (Erasable optical disks); [multisession burn]

* Magnetic Disk : A magnetic disk is a circular plate constructed of metal or plastic coated with magnetized material.

Both sides of the disk are used and several disks may be stacked on one spindle with read/write heads available on each surface.



Note: Hard drives use magnetic memory to store giga or terabytes of data in computer.

* Magnetic Tape. A magnetic tape transport consists of electric, mechanical and electronic components to provide the parts and control mechanism for a magnetic tape unit. The tape itself is a strip of plastic coated with a magnetic recording medium. Bits are recorded as magnetic spots on tape along several tracks. Seven or nine bits are recorded to form a character together with a Parity bit. R/W heads are mounted in each track so that data can be recorded and read as a sequence of characters.

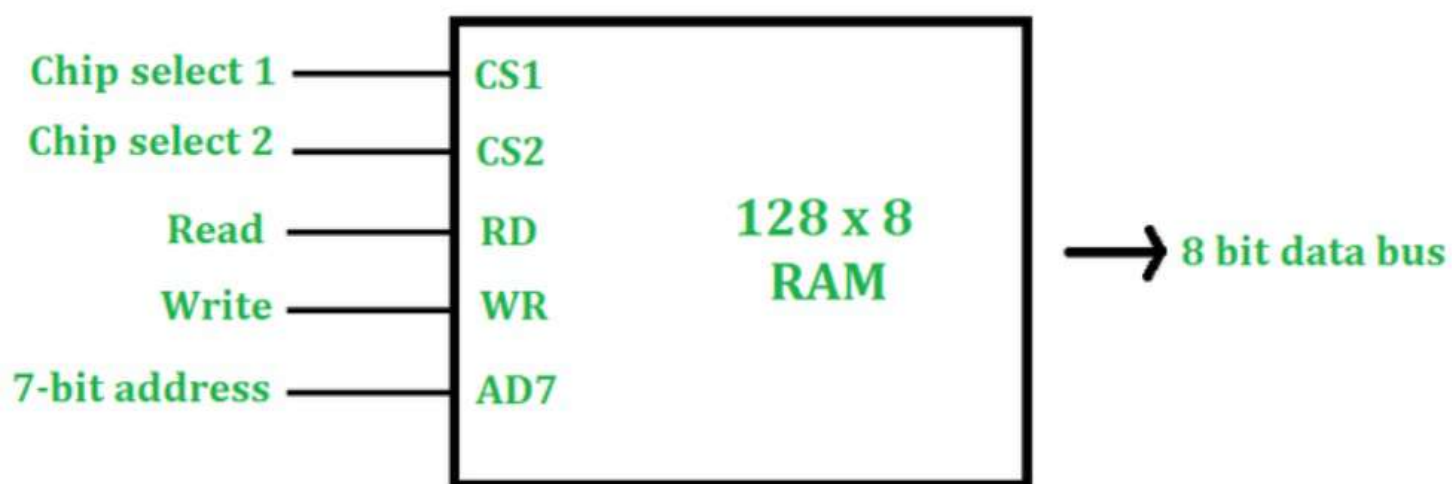
DVD → Digital versatile disc
Digital or video disc

CD → Compact Disc

Difference between Discs & Disk

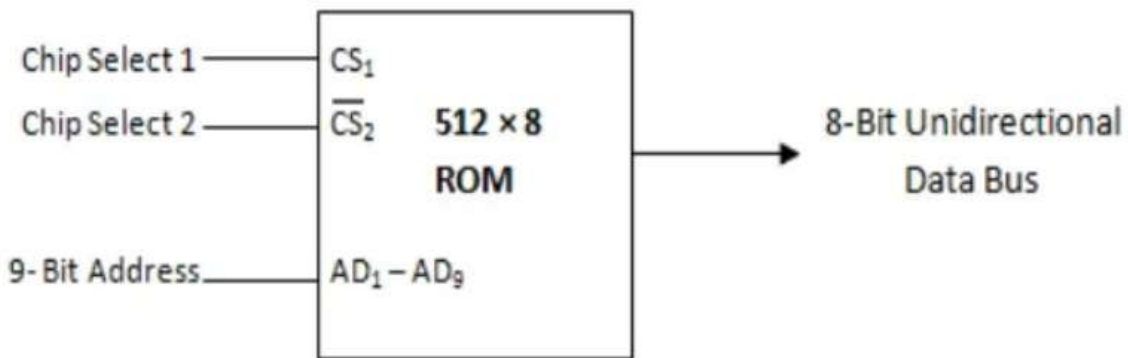
Disc {
⇒ the term disc refers to optical media such as an audio CD, CD-ROM, DVD-ROM, or DVD-video disc.
⇒ All discs are removable, meaning when we unmount or eject the disc from Desktop, it physically comes out of the computer.

⇒ A disk refers to magnetic media, such as floppy disk.
⇒ the disk in your computer's hard drive, an external hard drive.
⇒ Disks are always rewritable unless intentionally locked or write-protected. we can easily partition a disk into several small volume.
⇒ Disks are usually sealed inside a metal or plastic casing. Often, a disk and its enclosing mechanism are collectively known as "hard-drive".



Block diagram of typical RAM chip

ROM Chip



Static RAM	Dynamic RAM
➤ SRAM uses transistor to store a single bit of data	➤ DRAM uses a separate capacitor to store each bit of data
➤ SRAM does not need periodic refreshment to maintain data	➤ DRAM needs periodic refreshment to maintain the charge in the capacitors for data
➤ SRAM's structure is complex than DRAM	➤ DRAM's structure is simplex than SRAM
➤ SRAM are expensive as compared to DRAM	➤ DRAM's are less expensive as compared to SRAM
➤ SRAM are faster than DRAM	➤ DRAM's are slower than SRAM
➤ SRAM are used in Cache memory	➤ DRAM are used in Main memory

Point	RISC (Reduced Instruction Set Computer)	CISC (Complex Instruction Set Computer)
Instruction Set	Small and simple set of instructions	Large and complex instruction set
Execution Time	Most instructions execute in one clock cycle	Instructions may take multiple cycles
Instruction Format	Fixed length , simple	Variable length , complex
Hardware	Simple hardware, easy pipelining	Complex hardware, difficult pipelining
Code Size	Larger code (more instructions needed)	Smaller code (fewer instructions needed)
Examples	ARM, MIPS	Intel x86, AMD